Rheumatoid Arthritis

By

Mohammed Anter Ali

Lecturer of Orthopedic surgery

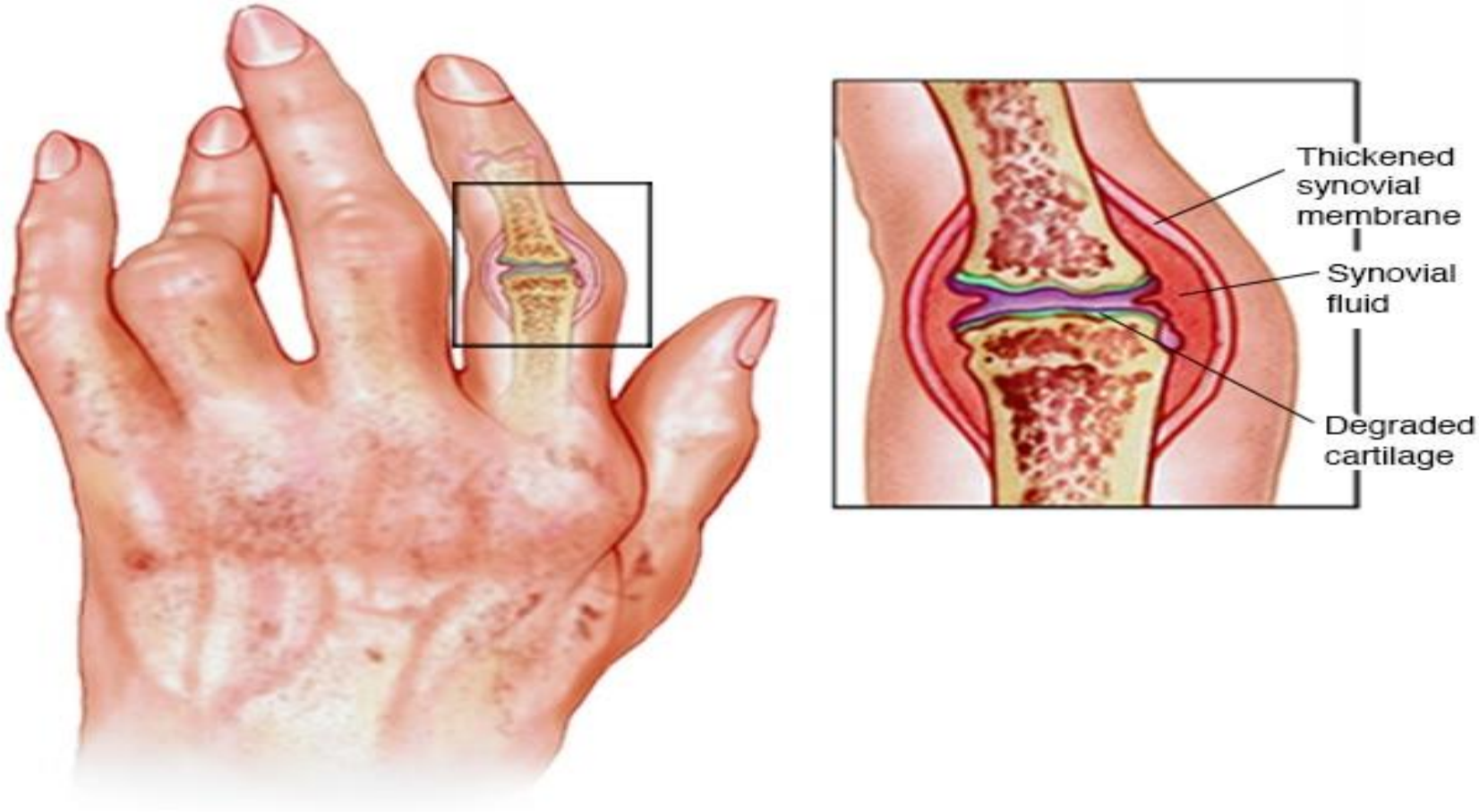
Content

- Definition
- Etiology
- Pathophysiology
- Clinical presentation
- Sign and symptoms
- Classification and progression
- RA at different joints
- Investigation
- Complication
- Management

Definition

it is an autoimmune disorder in which the joint lining and other tissues become inflamed as a result of over activity of body immune system.

RA is chronic, systemic, inflammatory disorder that primarily involves the joints.



© MAYO FOUNDATION FOR MEDICAL EDUCATION AND RESEARCH. ALL RIGHTS RESERVED.

Epidermiology

- **10 cases per 1000 people or 2.1 million adults in the united states**
- **Affects women two times more than males at all ages.**
- **There is general increase in prevalence for both gender with increasing age.**

Etiology

- **Like many other chronic disease the etiology of RA is unknown.**
- **A specific etiological agent for RA has not been identified.**
- **Based on the fact that individuals with RA produce antibodies to their own immunoglobulins , there is some reason to believe that RA is an autoimmune disorder**
- **It is not clear whether this antibody production is a primary event or result as a response to a specific antigen from an external stimulus.**

- **RHEMUATOID FACTOR-**

  RF have received considerable attention in search for a causative agent in RA because they are found in the serum of approximately 70% of all patients with RA.

Pathophysiology

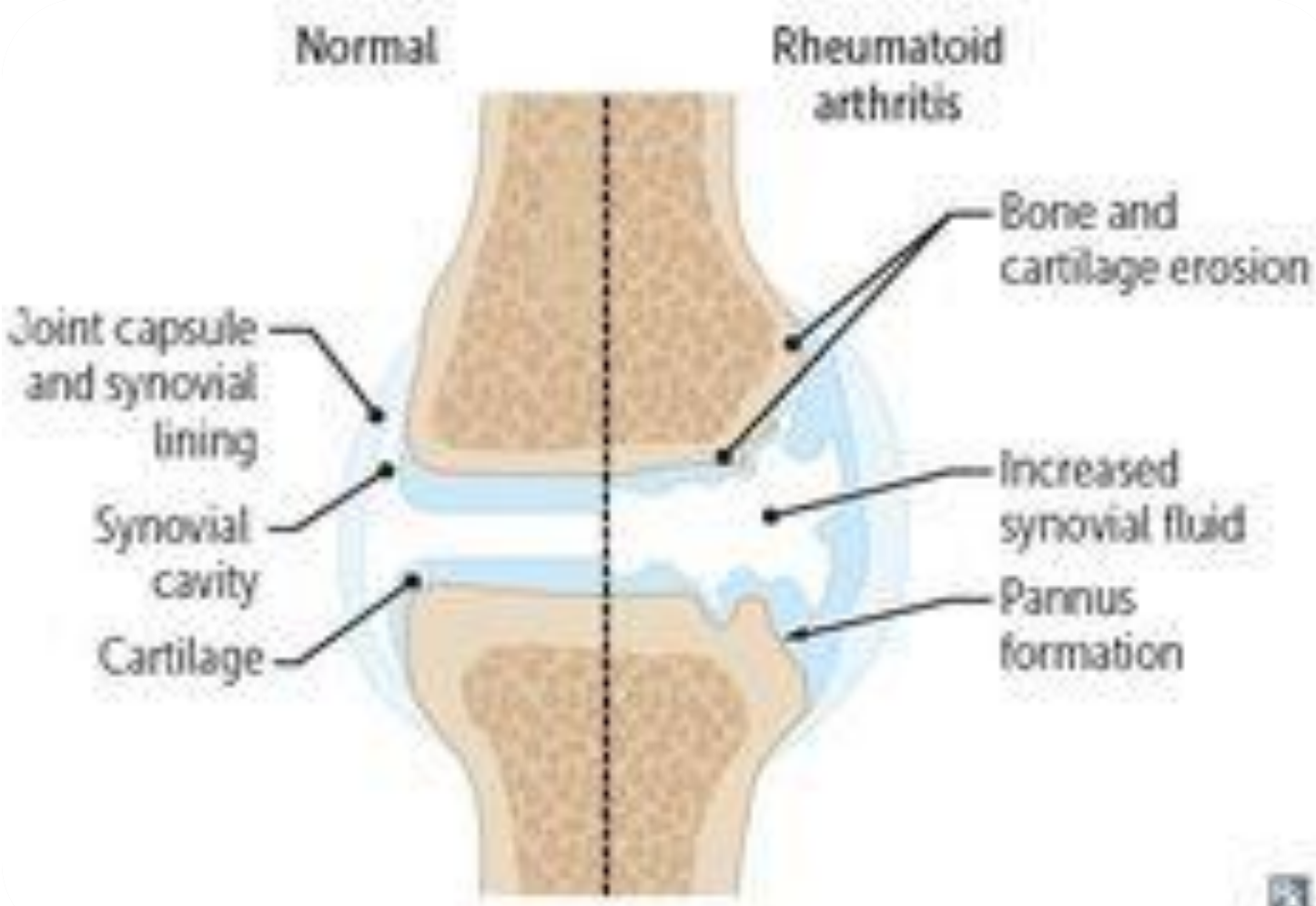
- **Chronic RA is characterized by the grossly edematous appearance of the synovium with hair like projections into the joint cavity.**
- **There are distinctive vascular changes, including venous distention, capillary obstruction, neutrophilic infiltration of the arterial walls, and areas of thrombosis and hemorrhage**

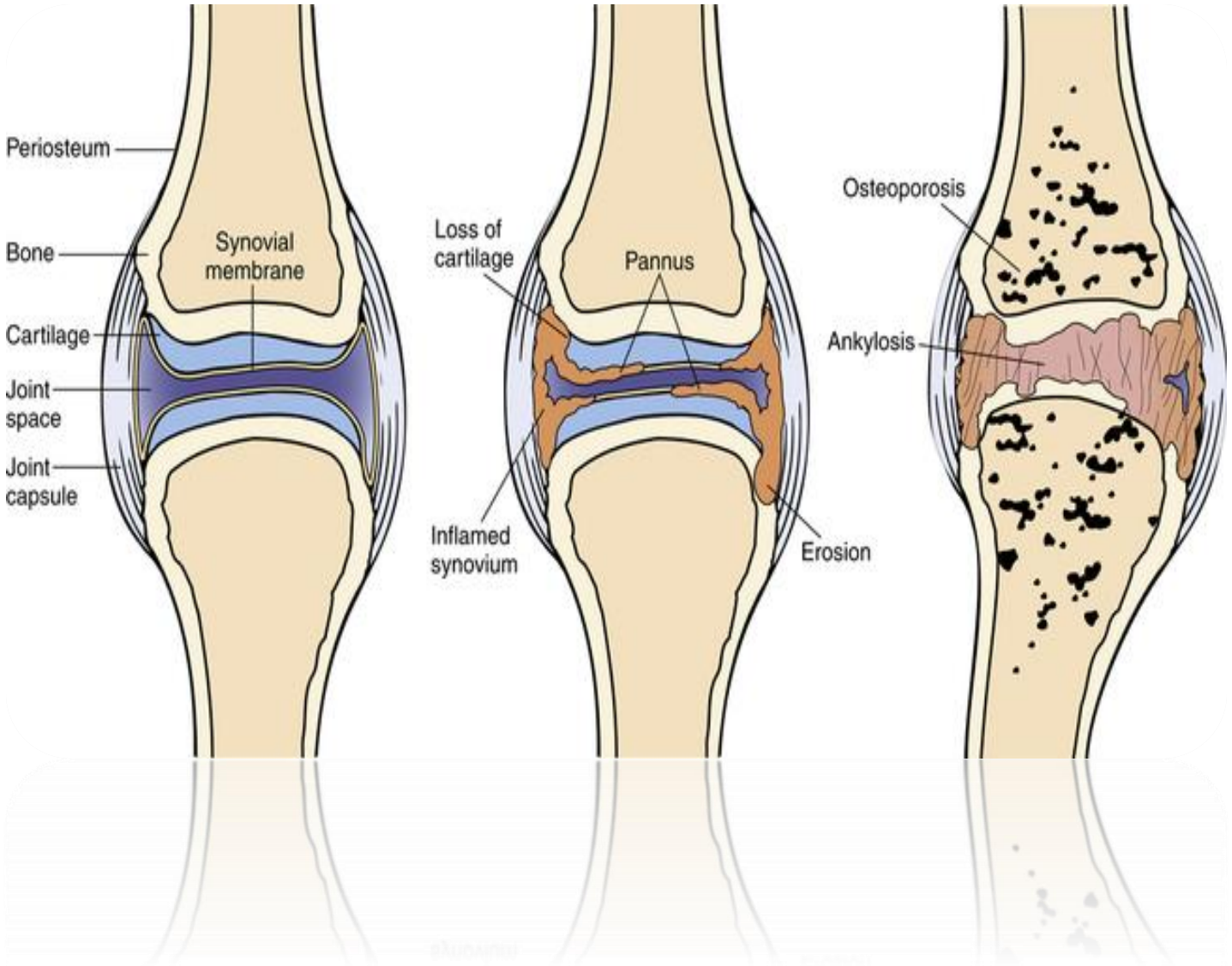
Conti…

- **Pannus - synovial proliferation of vascular granulation tissue, dissolves collagen as it extends over the joint cartilage. Granulation tissue will eventually result in adhesion, fibrosis or bony ankylosis of the joint.**

Conti…

- **Chronic inflammation can also weaken the joint capsule and it supporting ligamentous structures, altering structures and function.**
- **Tendon rupture and damaged tendon sheath may produce imbalance muscle pull resulting in deformities seen in advanced RA.**


Normal
Rheumatoid arthritis
Bone and cartilage erosion
Joint capsule and synovial lining
Increased synovial fluid
Synovial cavity
Pannus formation
Cartilage


Periosteum
Bone
Cartilage
Joint space
Joint capsule
Synovial membrane
Loss of cartilage
Pannus
Inflamed synovium
Erosion
Osteoporosis
Ankylosis

Clinical manifestations

1. **Morning stiffness**
2. **Arthritis of three or more joint areas- at least three joint areas simultaneously have had soft tissues swelling or fluid.**
3. **Arthritis of hand joints (wrist, MCP, PIP).**
4. **Symmetric arthritis**
5. **Rheumatoid nodules**

6. Serum rheumatoid factor

7. Radiographic changes- include erosion or unequivocal bony decalcification localized in or most marked adjacent to the involved joint.



Sings and symptoms

Know The Symptoms

Symptoms of Rheumatoid Arthritis

- Fatigue
- Morning Stiffness
- Joint Stiffness
- Joint Pain
- Minor Joint Swelling
- Fever
- Numbness and Tingling
- Decrease in Range of Motion



Classification and progression of rheumatoid arthritis

Stage – I Early

- **No destructive changes on radiographic examination**
- **Radiographic evidence of osteoporosis may be present.**

Stage –II Moderate

- **Radiographic evidence of osteoporosis with or without slight subchondral bone destruction , slight cartilage destruction may be present.**
- **No joint deformities ,but limitation of ROM may be present.**
- **Adjacent muscle atrophy**
- **Extra – articular soft tissue lesions such as nodules and tenosynovitis may be present**

Stage –III Severe

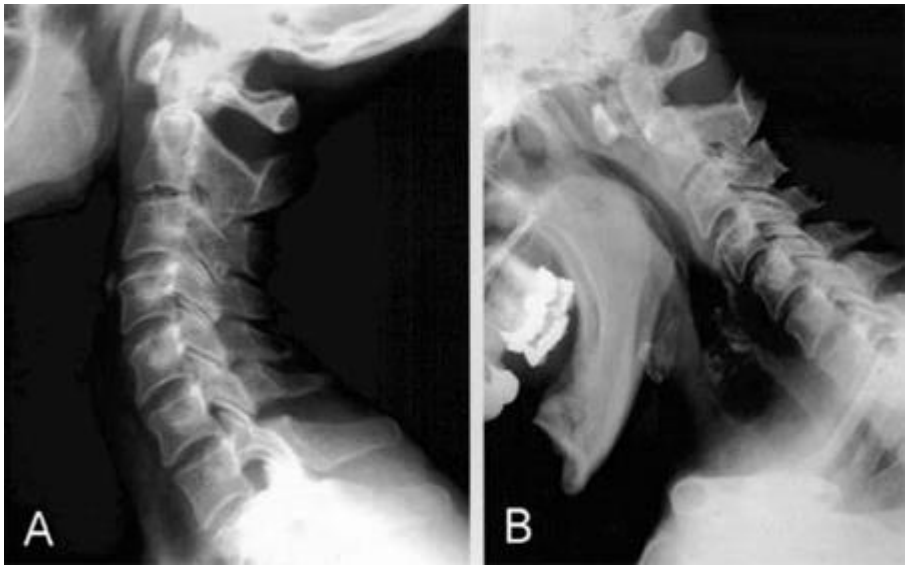
- **Radiographic evidence of cartilage and bone destruction in addition to osteoporosis.**
- **Joint deformity such as subluxation, ulnar deviation or hyperextention , without fibrous or bony ankylosis**
- **Extensive muscle atrophy**
- **Extra- articular soft tissue lesion such as nodules and tenosynovitis may present**

Stage –IV Terminal

- **Fibrous or bony ankylosis**
- **Criteria of stage 3**

Cervical spine

- **50 % takes place at the atlanto-axial joint.**
- **Ankylosing spondylitis**

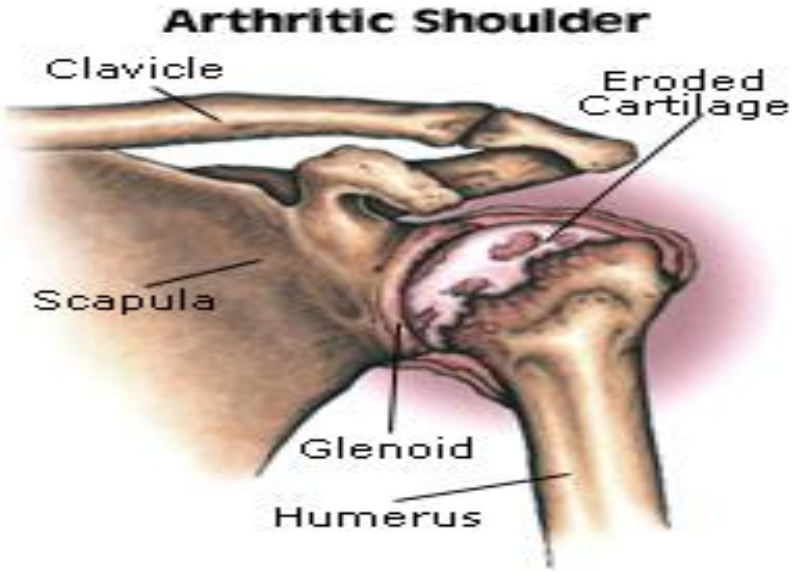




Shoulder

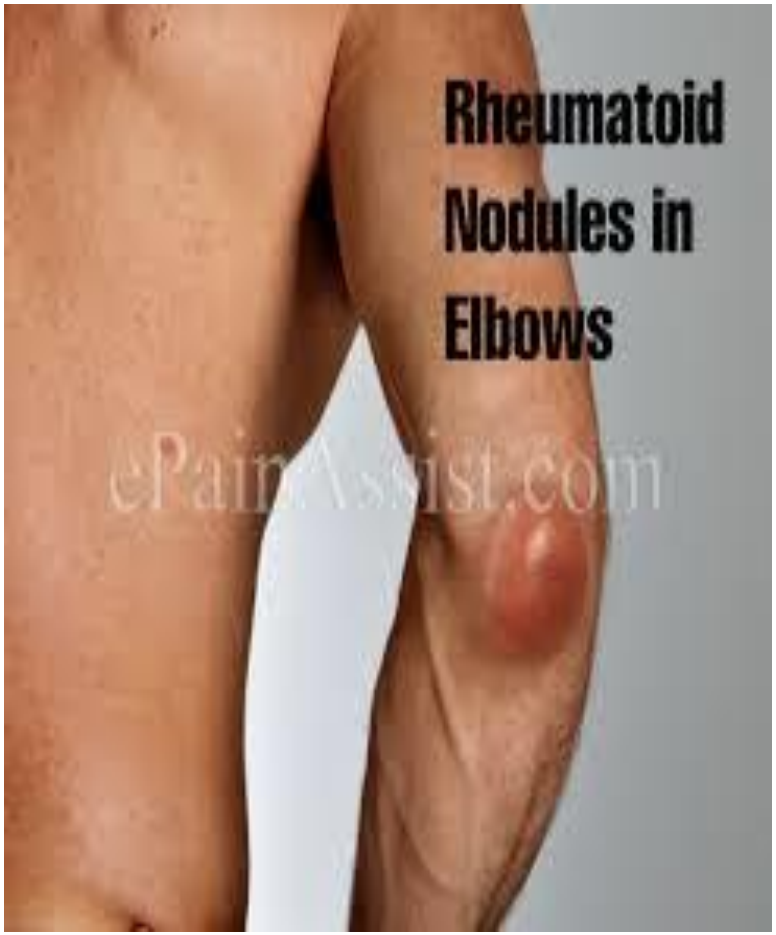
- **Involvement of GH, SC, AC joint and ST also**
- **Degeneration , pain, low ROM.**
- **Capsule and ligaments become distended with chronic inflamation**





Elbow

- **Capsular and ligamentous distention, and joint surface erosion may lead to elbow instability.**
- **Flexion contracture may result from persistent spasm secondary to pain.**



Wrist

- Development of flexion contractures which ultimately diminishes the ability to execute power grasp.
- Volar subluxation results from chronic inflammation of the proximal carpals
- Stenosing tenosynoitis may also occur

Rheumatoid Deformities at Wrist

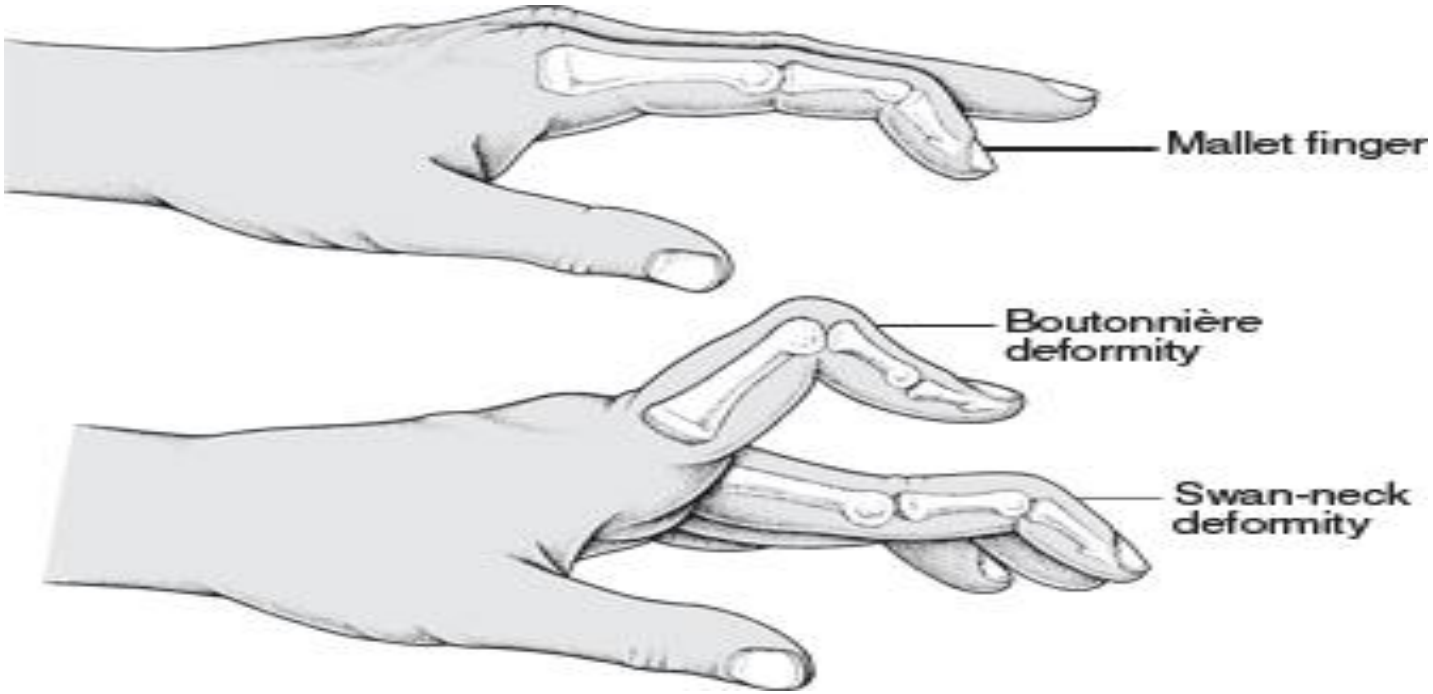


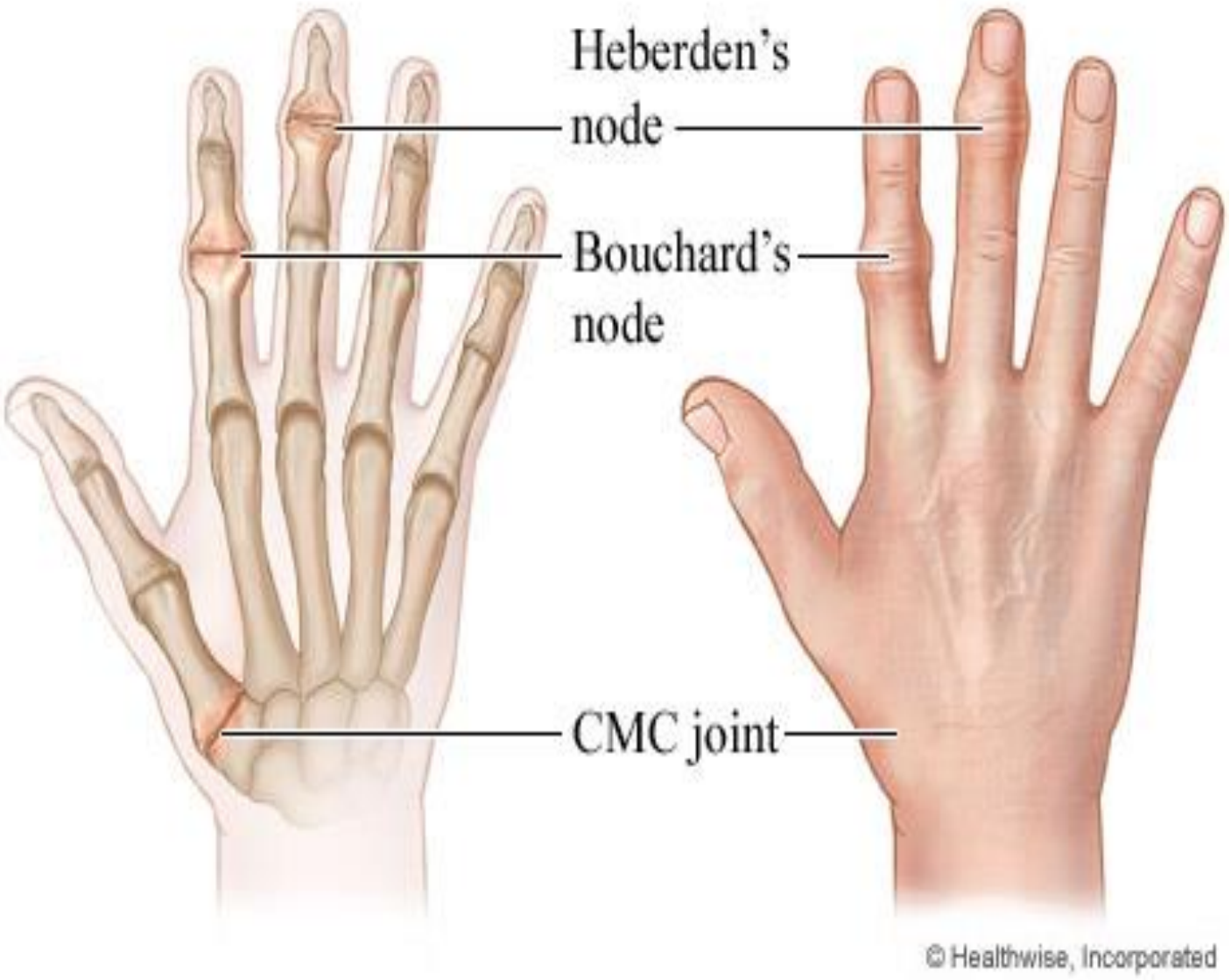



- Hourglass
- Caput Ulna
- Palmar subluxation and ulna drift
- Distal radioulnar joint dislocation

Hand

- **MCP joint- soft tissue swelling , volar subluxation , ulnar drift**
- **PIP joint – swelling (sausage like finger)**
  - **swan neck deformity**
  - **boutonniere deformity**
  - **bouchard's nodes**
- **DIP joint- heberden's nodes**
  - **mallet finger**


Mallet finger
Boutonnière deformity
Swan-neck deformity

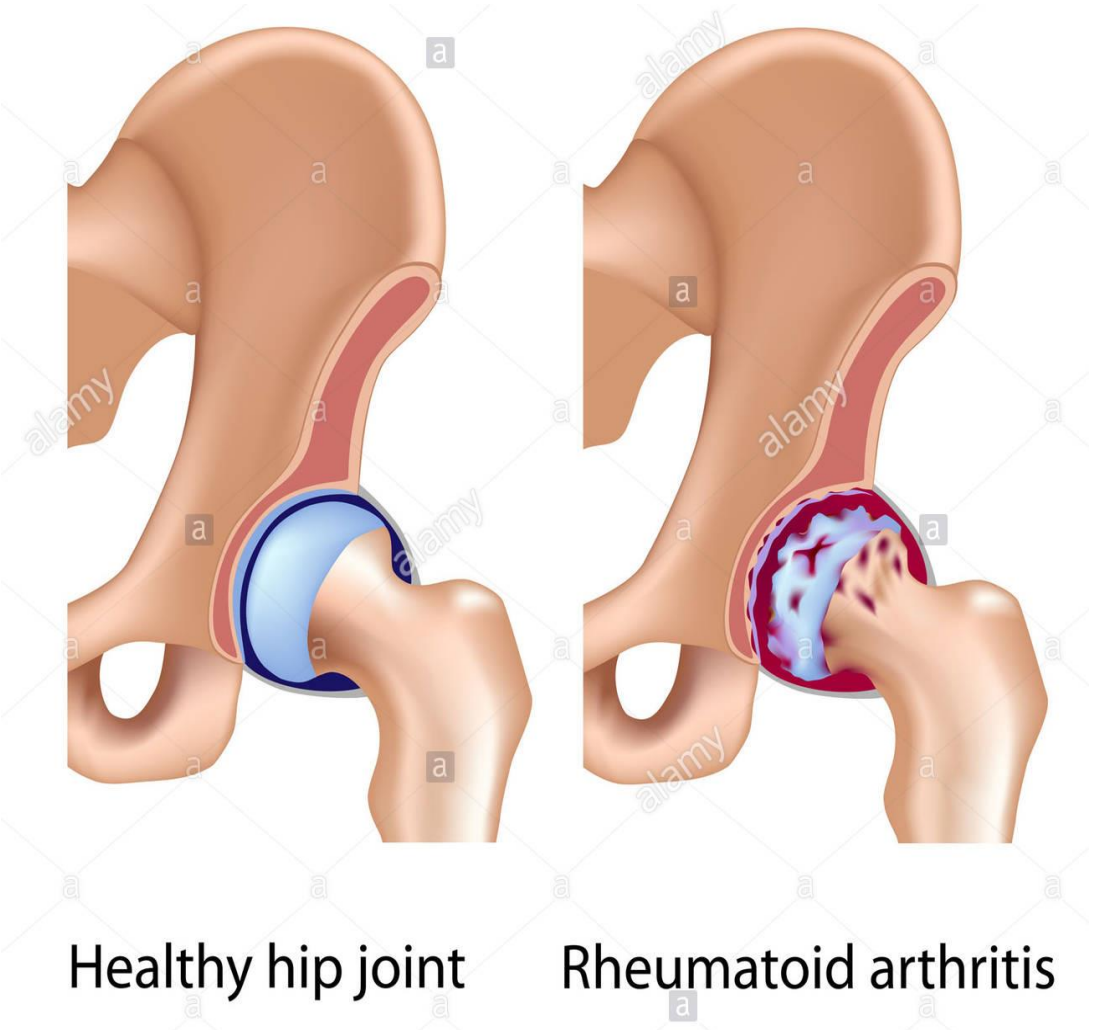

Heberden's node
Bouchard's node
CMC joint
© Healthwise, Incorporated

Thumb

- **Type 1 deformity- MCP flex, IP hyperextension without CMC involvement .**
- **Type II deformity- CMC is subluxed , IP hyperextension.**
- **Type III deformity- CMC subluxed , MCP hyperextension commonly found in RA.**

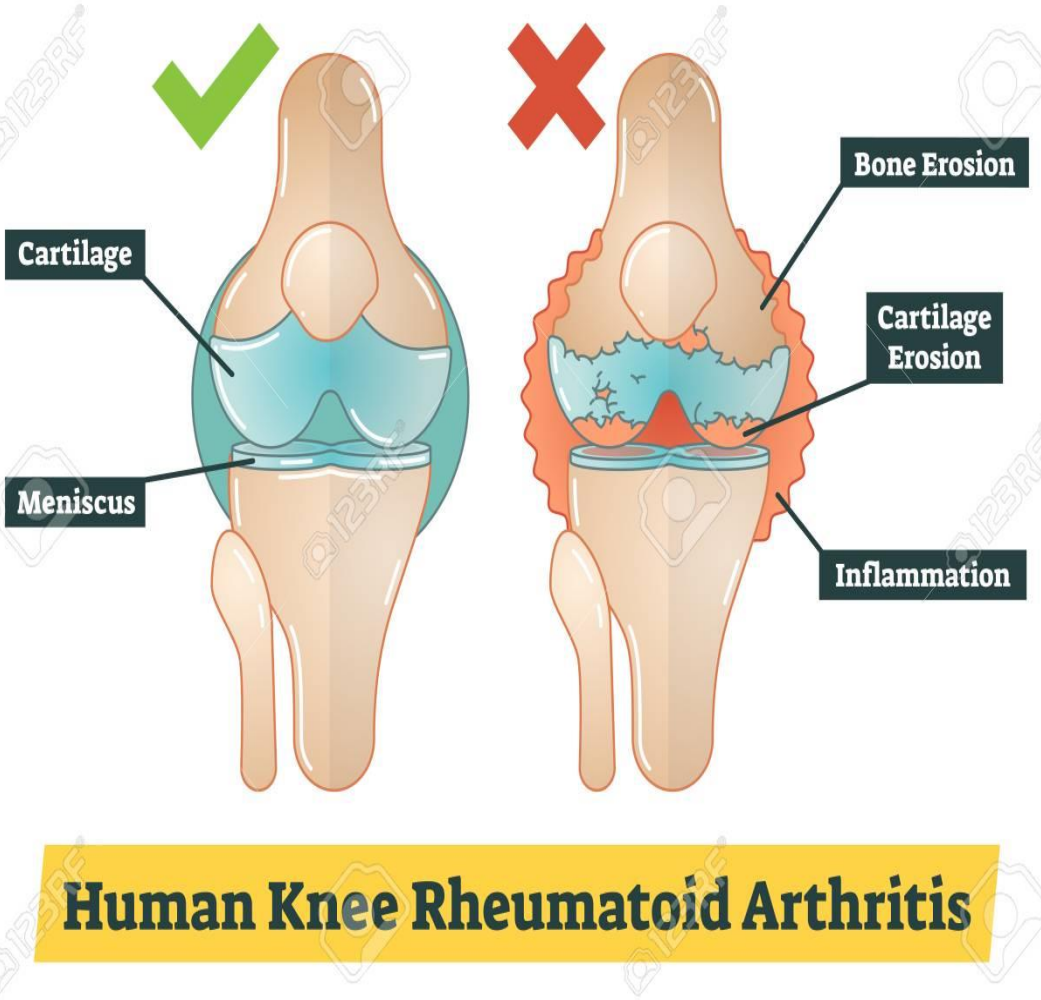
Hip

- Less commonly involved in RA
- Severe inflammatory destruction of the femoral head and the acetabulum into the pelvic cavity .



Knee

- Distention of the joint capsule and attenuation of ligaments.
- Painful knees may be held in slightly flexed positions, ultimately resulting to flexion contractures.

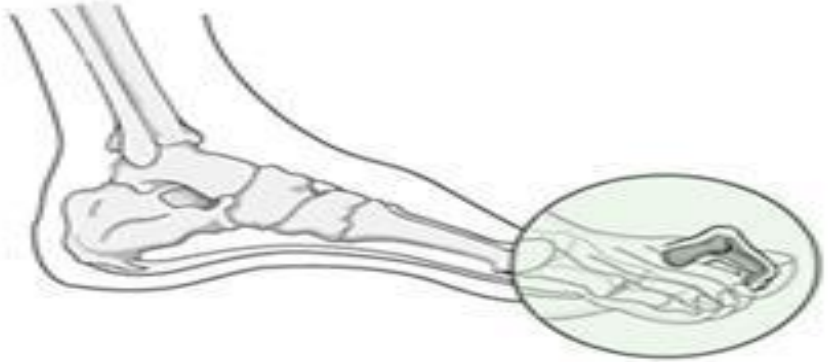


Ankle and feet

- Hindfoot pronation
- Flattening of the ML arch
- Calcaneal exostoses
- Splayfoot
- Metatarsalgia
- Hallux valgus and bunion
- Hammer toes
- Claw toes

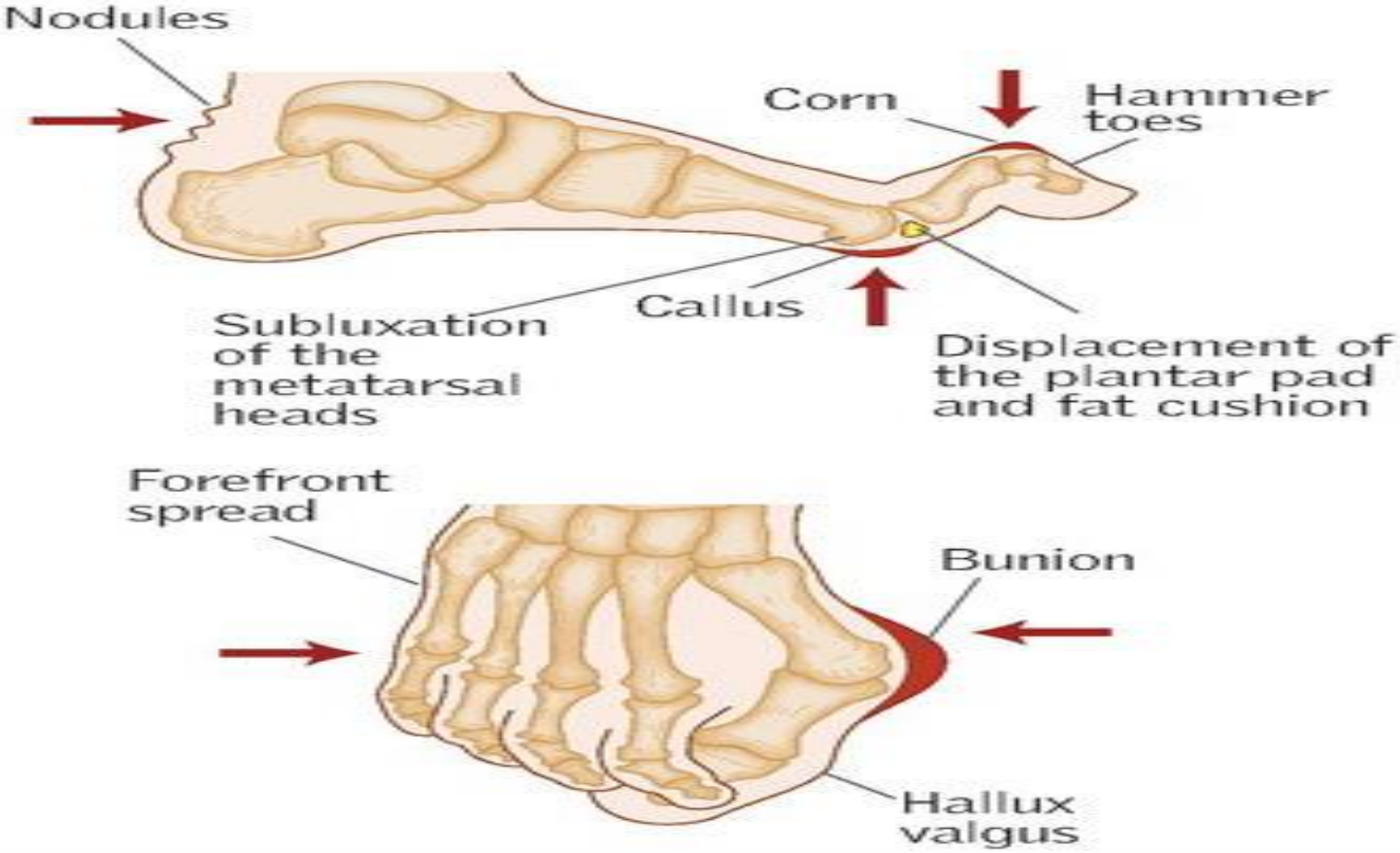


Bunion



Clawtoe

The arrows indicate areas at risk of ulceration



Investigations

- ESR and CRP
- Ultrasound or MRI
- RH factor
- X- rays
- Functional assessment
- CBC
- Urea, creatinine , liver function test

- **RHEUNATOID FACTOR – 85% of the patient with RA are RF positive**
- **RF (+) can be seen in other diseases such as rheumatic , viral, bacterial, parasitic, neoplasm .**

RHEUMATOID ARTHRITIS

Complications

- Rheumatoid nodules
- Cardiovascular: vasculitis, pericarditis
- Pulmonary: pulmonary nodules, interstitial fibrosis, pleuritis
- Ocular: scleritis, conjunctivitis
- Neurologic: compression neuropathy, vasculitis
- Skin: distal leg ulcers, palmar erythema
- Hematologic: granulocytopenia (Felty's syndrome), hyperviscosity
- Renal/hepatic: amyloidosis, ↑ liver enzymes

Harris ED Jr. In: Kelley WN et al, eds. *Textbook of Rheumatology*. 5th ed. Philadelphia, PA: WB Saunders; 1997:898–932.
Grassi W et al. *Eur J Radiol.* 1998;27(suppl 1):S18–S24.

Medical management

- **Medications – DMARD's , BRM's , corticosteroids, NSAIDs.**
- **Physiotherapy:**
- **Surgery:**
- **Synovectomy**
- **Arthrodesis Vs Arthroplasty**
- **Tendon reconstruction**

Physiotherapy management

- **GOALS-**
- relief of pain
- Prevention of deformities
- Correction of deformity
- Restoration & maintance of joint motion
- Improve muscle strength & endurance
- Gait training
- Education on management of re-occurance

Acute phase 3-4 weeks

- **Correct bed posture & supported positioing of involved joints.**
- Provide supported positioing of involved joints& correct bed posture
- Use firm mattress & back support to correct posture
- Limb is placed in position of minimal discomfort.

- **Additional support to limbs-** provide by splints and sandbags. Special attention is needed for the knee & elbow joint as they develop flexion contracture.
- **Improve vital capacity**
- **Improve ROM & Muscle strength-**isometrics, slow& relax isometrics
- **Functional mobility**
- **Postural guidance**
- **TENS, pulsed ultrasound**
- **Hydrotherapy**

Chronic phase

- **Thermotherapy**
- **Active and functional therapeutic program**
- **Stretching exercises**
- **Active and passive resisted exercises**
- **Gait training**

Equipment

- Adaptive for ADL
- Assistive for ambulation
- Appropriate footware or insoles

Education

- Self management
- Weight loss
- Activity management or joint protection
- Social support
- Stress management / relaxation

Splints



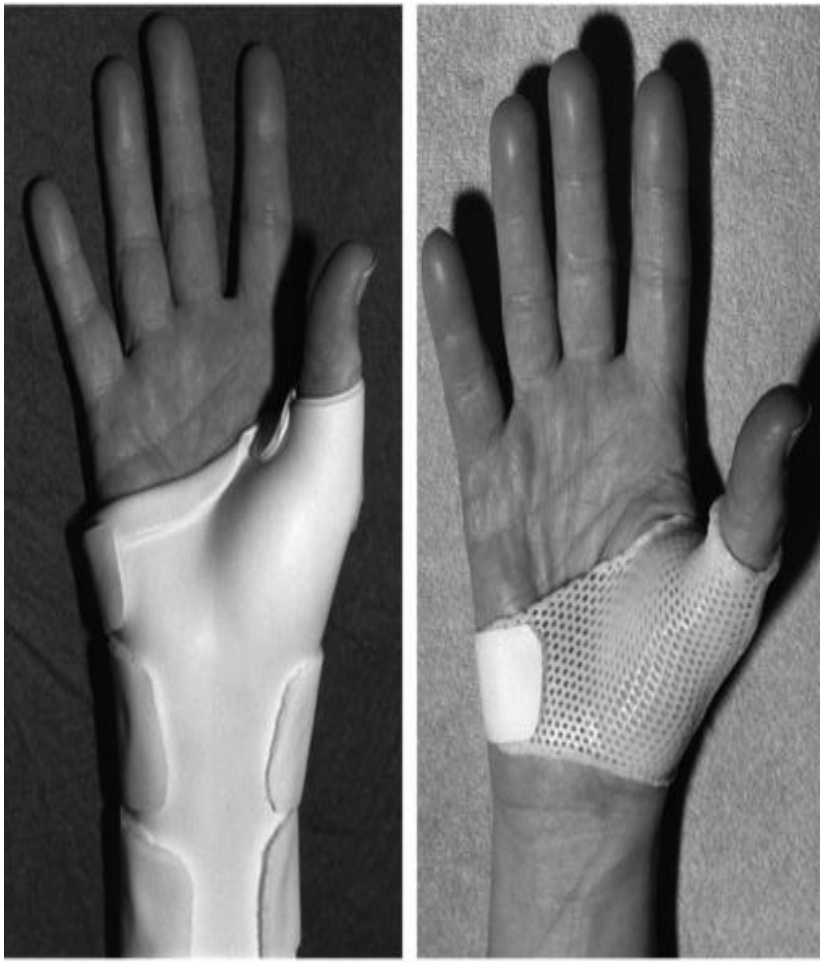
Knuckle Bender Splint for Claw Hand (Ulnar Nerve)



Cockup Splint for Wrist Drop (Radial Nerve)



Ankle Foot Orthosis for Foot Drop (Common Peroneal Nerve)



A B

Surgical management

- **Synovectomy**
- **Arthrodesis**
- **Arthroplasty**
- **Tendon reconstruction of Rupture tendons**

Thank you